

Assignment 14

Ellen Bledsoe

2026-10-06

2. When Did You Knit This Document? (5 pts)

```
[1] "2026-10-06"
```

```
[1] "2026-10-06 10:50:20 MST"
```

3. Plant Vouchers (20 pts)

```
Rows: 165 Columns: 17
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (11): season, sp_code, sci_name_fieldID, sci_name_profID, voucher, DNA, ...
```

```
dbl (6): year, month, day, easting, northing, elevation (m)
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# A tibble: 165 x 18
```

	year	month	day	collection_date	season	sp_code	sci_name_fieldID
	<dbl>	<dbl>	<dbl>	<date>	<chr>	<chr>	<chr>
1	2017	1	26	2017-01-26	<NA>	<NA>	Panicum miliaceum
2	2016	3	20	2016-03-20	winter	cass bauh	Cassia bauhinioides
3	2016	3	20	2016-03-20	winter	spha hast	Sphaeralcea coccinea
4	2016	3	20	2016-03-20	winter	amsi tess	Amsinckia tessellata
5	2016	3	20	2016-03-20	winter	micr lene	Uropappus lindleyi
6	2016	3	20	2016-03-20	winter	erig conc	Erigeron concinnus
7	2016	3	20	2016-03-20	winter	atri cane	Atriplex canescens
8	2016	3	20	2016-03-20	winter	euro lana	Eurotia lanata
9	2016	3	20	2016-03-20	winter	pros glan	Prosopis glandulosa

```

10 2016      3      20 2016-03-20      winter phac ariz Phacelia arizonica
# i 155 more rows
# i 11 more variables: sci_name_profID <chr>, voucher <chr>, DNA <chr>,
#   label_number <chr>, collector <chr>, location <chr>, easting <dbl>,
#   northing <dbl>, `elevation (m)` <dbl>, vial_barcode <chr>, notes <chr>

```

```
[1] "2016-03-20"
```

```
[1] "2019-04-01"
```

```
[1] "95644800s (~3.03 years)"
```

```

# A tibble: 165 x 19
   year month   day collection_date DOY season sp_code sci_name_fieldID
  <dbl> <dbl> <dbl> <date>          <dbl> <chr> <chr>    <chr>
1  2017     1    26 2017-01-26         26 <NA> <NA>    Panicum miliaceum
2  2016     3    20 2016-03-20         80 winter cass bauh Cassia bauhinoidea
3  2016     3    20 2016-03-20         80 winter spha hast Sphaeralcea coccinea
4  2016     3    20 2016-03-20         80 winter amsi tess Amsinckia tessellata
5  2016     3    20 2016-03-20         80 winter micr lene Uropappus lindleyi
6  2016     3    20 2016-03-20         80 winter erig conc Erigeron concinnus
7  2016     3    20 2016-03-20         80 winter atri cane Atriplex canescens
8  2016     3    20 2016-03-20         80 winter euro lana Eurotia lanata
9  2016     3    20 2016-03-20         80 winter pros glan Prosopis glandulosa
10 2016     3    20 2016-03-20         80 winter phac ariz Phacelia arizonica
# i 155 more rows
# i 11 more variables: sci_name_profID <chr>, voucher <chr>, DNA <chr>,
#   label_number <chr>, collector <chr>, location <chr>, easting <dbl>,
#   northing <dbl>, `elevation (m)` <dbl>, vial_barcode <chr>, notes <chr>

```

4. NDVI from the Santa Rita Experimental Range (20 pts)

```

# A tibble: 110,270 x 6
   datetime          r_mean g_mean b_mean ir_mean NDVI_c
  <chr>          <dbl> <dbl> <dbl> <dbl> <dbl>
1 2017-02-24 17:15:05     66    57    33    93 -0.0136
2 2017-02-24 17:30:05     67    56    31    92  0.0244
3 2017-02-24 17:45:06     72    58    30    91  0.0893
4 2017-02-24 18:00:05     77    58    28    87  0.179
5 2017-02-24 18:15:06     66    67    42    85  0.0668
6 2017-02-24 18:30:06     70    63    44    73 -0.792

```

```

7 2017-02-24 18:45:06      38      24      30      66 -0.0219
8 2017-02-24 19:00:06      21      12      15      19 -0.694
9 2017-02-25 06:15:05      24      14      19      19 -0.909
10 2017-02-25 06:30:05      48      37      43     104  0.127
# i 110,260 more rows

```

```
# A tibble: 110,270 x 6
```

```

  datetime      r_mean g_mean b_mean ir_mean NDVI_c
  <dtm>        <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
1 2017-02-24 17:15:05      66      57      33      93 -0.0136
2 2017-02-24 17:30:05      67      56      31      92  0.0244
3 2017-02-24 17:45:06      72      58      30      91  0.0893
4 2017-02-24 18:00:05      77      58      28      87  0.179
5 2017-02-24 18:15:06      66      67      42      85  0.0668
6 2017-02-24 18:30:06      70      63      44      73 -0.792
7 2017-02-24 18:45:06      38      24      30      66 -0.0219
8 2017-02-24 19:00:06      21      12      15      19 -0.694
9 2017-02-25 06:15:05      24      14      19      19 -0.909
10 2017-02-25 06:30:05      48      37      43     104  0.127
# i 110,260 more rows

```

```
[1] "220585501s (~6.99 years)"
```

```
# A tibble: 110,270 x 9
```

```

  datetime      r_mean g_mean b_mean ir_mean NDVI_c year month DOY
  <dtm>        <dbl>  <dbl>  <dbl>  <dbl>  <dbl> <dbl> <dbl> <dbl>
1 2017-02-24 17:15:05      66      57      33      93 -0.0136 2017     2    55
2 2017-02-24 17:30:05      67      56      31      92  0.0244 2017     2    55
3 2017-02-24 17:45:06      72      58      30      91  0.0893 2017     2    55
4 2017-02-24 18:00:05      77      58      28      87  0.179 2017     2    55
5 2017-02-24 18:15:06      66      67      42      85  0.0668 2017     2    55
6 2017-02-24 18:30:06      70      63      44      73 -0.792 2017     2    55
7 2017-02-24 18:45:06      38      24      30      66 -0.0219 2017     2    55
8 2017-02-24 19:00:06      21      12      15      19 -0.694 2017     2    55
9 2017-02-25 06:15:05      24      14      19      19 -0.909 2017     2    56
10 2017-02-25 06:30:05      48      37      43     104  0.127 2017     2    56
# i 110,260 more rows

```

```

`summarise()` has regrouped the output.
i Summaries were computed grouped by year and month.
i Output is grouped by year.

```

```
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(year, month))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 85 x 3
# Groups:   year [8]
   year month mean_NVDI
  <dbl> <dbl>   <dbl>
1  2017     2 -0.0907
2  2017     3 -0.0802
3  2017     4 -0.0739
4  2017     5 -0.0734
5  2017     6 -0.0976
6  2017     7 -0.0810
7  2017     8  0.00581
8  2017     9 -0.0542
9  2017    10 -0.105
10 2017    11 -0.129
# i 75 more rows
```

5. Vectors (10 pts)

```
[1] "2a:"
```

```
[1] 23 24 40
```

```
[1] "2b:"
```

```
[1] TRUE FALSE FALSE
```

```
[1] "2c:"
```

```
[1] 0 0 4
```

```
[1] "2d:"
```

```
[1] "(woo)" "(hey)" NA
```

6. Dugout Data (15 pts)

Rows: 102 Columns: 16

-- Column specification -----

Delimiter: ",",

chr (8): Site_ID, Date, Soil Salinity, pH, Soil Zone, Location of nearest o...

dbl (7): latitude, longitude, Elevation.m, ion Concentration in groundwater...

time (1): Time

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

A tibble: 102 x 16

	Site_ID	Date	Time	latitude	longitude	SoilSalinity	pH	SoilZone
	<chr>	<chr>	<time>	<dbl>	<dbl>	<chr>	<chr>	<chr>
1	5	24-Aug-17	10:03	51.4	-103.	moderate	alkaline	dark gr~
2	20	24-Jul-17	11:41	50.1	-102.	very slight	unclassifi~	black
3	36	10-Aug-17	15:05	52.5	-105.	very slight	alkaline	dark gr~
4	49	24-Jul-17	13:15	50.0	-102.	slight	unclassifi~	black
5	51	24-Jul-17	16:19	50.0	-102.	slight	unclassifi~	black
6	52	25-Jul-17	11:27	49.9	-102.	slight	unclassifi~	black
7	65	11-Aug-17	11:50	52.6	-110.	very slight	slightly a~	dark br~
8	68	8-Aug-17	09:30	50.6	-105.	very slight	alkaline	brown
9	10A	24-Aug-17	12:25	51.8	-103.	slight	alkaline	dark gr~
10	10B	24-Aug-17	13:14	51.8	-103.	slight	alkaline	dark gr~

i 92 more rows

i 8 more variables: Elevation.m <dbl>,

`Location of nearest observation well` <chr>,

`ion Concentration in groundwater (mg/L)` <dbl>, MajorSalts <chr>,

Anion <chr>, `2017 Well groundwater depth` <dbl>,

`dugout elevation above groundwater` <dbl>, Surface_Sal.ppt <dbl>

[1] "3a:"

A tibble: 85 x 16

	Site_ID	Date	Time	latitude	longitude	SoilSalinity	pH	SoilZone
	<chr>	<chr>	<time>	<dbl>	<dbl>	<chr>	<chr>	<chr>
1	20	24-Jul-17	11:41	50.1	-102.	very slight	unclassifi~	black
2	36	10-Aug-17	15:05	52.5	-105.	very slight	alkaline	dark gr~
3	49	24-Jul-17	13:15	50.0	-102.	slight	unclassifi~	black
4	51	24-Jul-17	16:19	50.0	-102.	slight	unclassifi~	black
5	52	25-Jul-17	11:27	49.9	-102.	slight	unclassifi~	black

```

6 65      11-Aug-17 11:50      52.6      -110. very slight  slightly a~ dark br~
7 68      8-Aug-17 09:30      50.6      -105. very slight  alkaline   brown
8 10A     24-Aug-17 12:25      51.8      -103. slight       alkaline   dark gr~
9 10B     24-Aug-17 13:14      51.8      -103. slight       alkaline   dark gr~
10 10C    24-Aug-17 10:30      51.8      103. very slight   alkaline   dark gr~

```

```
# i 75 more rows
```

```
# i 8 more variables: Elevation.m <dbl>,
```

```
# `Location of nearest observation well` <chr>,
```

```
# `ion Concentration in groundwater (mg/L)` <dbl>, MajorSalts <chr>,
```

```
# Anion <chr>, `2017 Well groundwater depth` <dbl>,
```

```
# `dugout elevation above groundwater` <dbl>, Surface_Sal.ppt <dbl>
```

```
[1] "3b:"
```

```
# A tibble: 94 x 16
```

```

  Site_ID Date      Time    latitude longitude SoilSalinity pH      SoilZone
  <chr>   <chr>    <time>    <dbl>    <dbl> <chr>      <chr>    <chr>
1 10A     24-Aug-17 12:25      51.8     -103. slight      alkaline   dark gr~
2 10B     24-Aug-17 13:14      51.8     -103. slight      alkaline   dark gr~
3 10C     24-Aug-17 10:30      51.8      103. very slight   alkaline   dark gr~
4 10D     24-Aug-17 11:39      51.8     -103. very slight   alkaline   dark gr~
5 14A     12-Jul-17 10:15      51.0     -105. very slight   alkaline   brown
6 14B     12-Jul-17 12:50      51.0     -105. very slight   alkaline   black
7 15A     3-Aug-17  11:41      49.6     -102. slight       neutral to~ dark gr~
8 15B     3-Aug-17  14:15      49.5     -102. slight       neutral to~ dark gr~
9 22B     8-Aug-17  12:28      51.1      106. very slight   alkaline   brown
10 24A    14-Aug-17 14:15      49.9     -110. slight       neutral to~ brown

```

```
# i 84 more rows
```

```
# i 8 more variables: Elevation.m <dbl>,
```

```
# `Location of nearest observation well` <chr>,
```

```
# `ion Concentration in groundwater (mg/L)` <dbl>, MajorSalts <chr>,
```

```
# Anion <chr>, `2017 Well groundwater depth` <dbl>,
```

```
# `dugout elevation above groundwater` <dbl>, Surface_Sal.ppt <dbl>
```

```
[1] "3c:"
```

```
# A tibble: 102 x 16
```

```

  Site_ID Date      Time    latitude longitude SoilSalinity pH      SoilZone
  <chr>   <chr>    <time>    <dbl>    <dbl> <chr>      <chr>    <chr>
1 5       24-Aug-17 10:03      51.4     -103. moderate      alkaline   dark gr~
2 20      24-Jul-17 11:41      50.1     -102. very slight   unclassifi~ black

```

```

3 36      10-Aug-17 15:05      52.5      -105. very slight  alkaline      dark gr~
4 49      24-Jul-17 13:15      50.0      -102. slight      unclassifi~ black
5 51      24-Jul-17 16:19      50.0      -102. slight      unclassifi~ black
6 52      25-Jul-17 11:27      49.9      -102. slight      unclassifi~ black
7 65      11-Aug-17 11:50      52.6      -110. very slight  slightly a~ dark br~
8 68      8-Aug-17 09:30      50.6      -105. very slight  alkaline      brown
9 10A     24-Aug-17 12:25      51.8      -103. slight      alkaline      dark gr~
10 10B    24-Aug-17 13:14      51.8      -103. slight      alkaline      dark gr~
# i 92 more rows
# i 8 more variables: Elevation.m <dbl>,
#   `Location of nearest observation well` <chr>,
#   `ion Concentration in groundwater (mg/L)` <dbl>, MajorSalts <chr>,
#   Anion <chr>, `2017 Well groundwater depth` <dbl>,
#   `dugout elevation above groundwater` <dbl>, Surface_Sal.ppt <dbl>

```

7. Santa Cruz Rodents (20 pts)

Rows: 51 Columns: 15

```

-- Column specification -----
Delimiter: ", "
chr  (10): Site, Trap ID, Species, Status (R/N), Sex, Tail length, Hair samp...
dbl  (4): Total Weight, Bag weight, Animal Weight, Hind foot length
date (1): Date

```

```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

```
[1] "Read in data:"
```

```
[1] "4a:"
```

```

# A tibble: 2 x 15
  Date       Site `Trap ID` Species `Status (R/N)` Sex   `Total Weight`
  <date>     <chr> <chr>    <chr>   <chr>         <chr>         <dbl>
1 2022-11-14 <NA>  4J      SIOC?   N             <NA>          NA
2 2022-11-18 <NA>  D6      DIME?   N             F             44
# i 8 more variables: `Bag weight` <dbl>, `Animal Weight` <dbl>,
#   `Hind foot length` <dbl>, TailLength <chr>, HairSample <chr>,
#   Position <chr>, Handler <chr>, Notes <chr>

```

```
[1] "4b:"
```

```
# A tibble: 51 x 15
  Date       Site   `Trap ID` Species `Status (R/N)` Sex   `Total Weight`
  <date>     <chr>   <chr>     <chr>   <chr>         <chr>         <dbl>
1 2022-11-14 Heritage 4C       SIOC     N           F           134
2 2022-11-14 <NA>    4D       SIOC     N           M           136
3 2022-11-14 <NA>    4I       SIOC     N           <NA>        90
4 2022-11-14 <NA>    2H       REME     N           M           38
5 2022-11-14 <NA>    4J       SIOC?    N           <NA>        NA
6 2022-11-14 <NA>    2F       REME     N           F           22
7 2022-11-15 <NA>    4C       SIOC     R           <NA>        NA
8 2022-11-15 <NA>    4H       SIOC     N           F           95
9 2022-11-15 <NA>    1H       REME     N           <NA>        26
10 2022-11-15 <NA>    1B       REME     N           F           35
# i 41 more rows
# i 8 more variables: `Bag weight` <dbl>, `Animal Weight` <dbl>,
#   `Hind foot length` <dbl>, TailLength <chr>, HairSample <chr>,
#   Position <chr>, Handler <chr>, Notes <chr>
```

```
[1] "4c"
```

```
# A tibble: 51 x 15
  Date       Site   `Trap ID` Species `Status (R/N)` Sex   `Total Weight`
  <date>     <chr>   <chr>     <chr>   <chr>         <chr>         <dbl>
1 2022-11-14 Heritage 4C       SIOC     N           F           134
2 2022-11-14 <NA>    4D       SIOC     N           M           136
3 2022-11-14 <NA>    4I       SIOC     N           <NA>        90
4 2022-11-14 <NA>    2H       REME     N           M           38
5 2022-11-14 <NA>    4J       SIOC     N           <NA>        NA
6 2022-11-14 <NA>    2F       REME     N           F           22
7 2022-11-15 <NA>    4C       SIOC     R           <NA>        NA
8 2022-11-15 <NA>    4H       SIOC     N           F           95
9 2022-11-15 <NA>    1H       REME     N           <NA>        26
10 2022-11-15 <NA>    1B       REME     N           F           35
# i 41 more rows
# i 8 more variables: `Bag weight` <dbl>, `Animal Weight` <dbl>,
#   `Hind foot length` <dbl>, TailLength <chr>, HairSample <chr>,
#   Position <chr>, Handler <chr>, Notes <chr>
```

```
[1] "4d:"
```

```
# A tibble: 51 x 15
```

	Date	Site	Trap ID	Species	Status (R/N)	Sex	Total Weight
	<date>	<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>
1	2022-11-14	Heritage	4C	SIOC	N	F	134
2	2022-11-14	<NA>	4D	SIOC	N	M	136
3	2022-11-14	<NA>	4I	SIOC	N	<NA>	90
4	2022-11-14	<NA>	2H	REME	N	M	38
5	2022-11-14	<NA>	4J	SIOC	N	<NA>	NA
6	2022-11-14	<NA>	2F	REME	N	F	22
7	2022-11-15	<NA>	4C	SIOC	R	<NA>	NA
8	2022-11-15	<NA>	4H	SIOC	N	F	95
9	2022-11-15	<NA>	1H	REME	N	<NA>	26
10	2022-11-15	<NA>	1B	REME	N	F	35

i 41 more rows
i 8 more variables: `Bag weight` <dbl>, `Animal Weight` <dbl>,
`Hind foot length` <dbl>, TailLength <chr>, HairSample <chr>,
Position <chr>, Handler <chr>, Notes <chr>

8. Challenge: What Do Kangaroo Rats Eat? (Optional)

```
Rows: 299 Columns: 12
-- Column specification -----
Delimiter: ","
chr (6): vial_barcode, species, sex, sample_type, PIT_tag, notes
dbl (6): month, day, year, period, plot, stake

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 24101 Columns: 19
-- Column specification -----
Delimiter: ","
chr (11): OTU, SampleID, DataFrame, ConsensusLineage, Domain, Clade1, Class,...
dbl (8): Reads, WTU.domain, WTU.clade1, WTU.class, WTU.order, WTU.family, W...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

8a.

```
# A tibble: 299 x 14
  month   day year period vial_barcode plot stake species sex sample_type
  <dbl> <dbl> <dbl> <dbl> <chr>         <dbl> <dbl> <chr>   <chr> <chr>
```

1	10	1	2016	454	S009076	4	22	D0	M	fresh
2	10	1	2016	454	S009095	4	22	D0	M	trap
3	10	1	2016	454	S009066	4	64	DM	F	trap
4	10	1	2016	454	S009067	4	43	D0	F	fresh
5	10	1	2016	454	S009065	4	43	D0	F	trap
6	10	1	2016	454	S009090	4	21	DM	F	trap
7	10	1	2016	454	S009097	11	73	DM	M	trap
8	10	1	2016	454	S009086	11	61	PP	F	trap
9	10	1	2016	454	S009096	11	75	DM	M	trap
10	10	1	2016	454	S009074	11	45	DM	F	fresh

i 289 more rows
i 4 more variables: PIT_tag <chr>, notes <chr>, collection_date <date>,
month_name <ord>

8b.

A tibble: 14,827 x 4

	SampleID	OTU	Reads	ConsensusLineage
	<chr>	<chr>	<dbl>	<chr>
1	S008813	OTU5	3896	k__Eukaryota; p__Streptophyta; c__None; o__Brassicale~
2	S008813	OTU22	574	k__Eukaryota; p__Streptophyta; c__None; o__Geraniales~
3	S008813	OTU187	212	k__Eukaryota; p__Streptophyta; c__None; o__Brassicale~
4	S008813	OTU1296	41	k__Eukaryota; p__Streptophyta; c__None; o__Brassicale~
5	S008813	OTU180	37	k__Eukaryota; p__Streptophyta; c__None; o__Geraniales~
6	S008813	OTU1050	32	k__Eukaryota; p__Streptophyta; c__None; o__Brassicale~
7	S008813	OTU139	24	k__Eukaryota; p__Streptophyta; c__None; o__Geraniales~
8	S008813	OTU1866	18	k__Eukaryota; p__Streptophyta; c__None; o__Brassicale~
9	S008813	OTU39	14	k__Eukaryota; p__Streptophyta; c__None; o__Boraginale~
10	S008813	OTU18	11	k__Eukaryota; p__Streptophyta; c__None; o__Caryophyll~

i 14,817 more rows

8c.

A tibble: 14,824 x 5

	SampleID	OTU	Reads	family	ConsensusLineage
	<chr>	<chr>	<dbl>	<chr>	<chr>
1	S008813	OTU5	3896	Brassicaceae	k__Eukaryota; p__Streptophyta; c__Non~
2	S008813	OTU22	574	Geraniaceae	k__Eukaryota; p__Streptophyta; c__Non~
3	S008813	OTU187	212	Brassicaceae	k__Eukaryota; p__Streptophyta; c__Non~
4	S008813	OTU1296	41	Brassicaceae	k__Eukaryota; p__Streptophyta; c__Non~
5	S008813	OTU180	37	Geraniaceae	k__Eukaryota; p__Streptophyta; c__Non~

```

6 S008813 OTU1050 32 Brassicaceae k__Eukaryota; p__Streptophyta; c__Non~
7 S008813 OTU139 24 Geraniaceae k__Eukaryota; p__Streptophyta; c__Non~
8 S008813 OTU1866 18 Brassicaceae k__Eukaryota; p__Streptophyta; c__Non~
9 S008813 OTU39 14 Hydrophyllaceae k__Eukaryota; p__Streptophyta; c__Non~
10 S008813 OTU18 11 Polygonaceae k__Eukaryota; p__Streptophyta; c__Non~
# i 14,814 more rows

```

8d.

```

# A tibble: 71 x 4
  month_name family      total_reads prop_reads
  <ord>      <chr>          <dbl>      <dbl>
1 Mar      Aizoaceae             52  0.000436
2 Mar      Amaranthaceae        140  0.00117
3 Mar      Asteraceae          1578  0.0132
4 Mar      Boraginaceae          1617  0.0136
5 Mar      Brassicaceae         49365  0.414
6 Mar      Chenopodiaceae        1000  0.00839
7 Mar      Cupressaceae           3  0.0000252
8 Mar      Fabaceae            22469  0.188
9 Mar      Fagaceae             21  0.000176
10 Mar     Geraniaceae          13561  0.114
# i 61 more rows

```

8e.

```

# A tibble: 39 x 4
  family      Mar      Sep      Oct
  <chr>      <dbl>  <dbl>  <dbl>
1 Brassicaceae 0.414  0.449  0.0298
2 Poaceae      0.191  0.105  0.301
3 Fabaceae     0.188  0.0532 0.192
4 Geraniaceae  0.114  0.00103 0.00000630
5 Hydrophyllaceae 0.0459 0      0
6 Boraginaceae 0.0136 0      0.0000189
7 Asteraceae   0.0132 0.00586 0.215
8 Chenopodiaceae 0.00839 0.00497 0.000183
9 Polemoniaceae 0.00464 0      0
10 Plantaginaceae 0.00141 0.0300 0.0000126
# i 29 more rows

```

8f.

